



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 04

Team No.: 19

Project Title: Debugging

Team Members

Roll Number	Student Name	Signature
2520030418	A.Charan Teja	
2520030419	S. Teja	
2520030475	E. Ashwith Reddy	

1. Abstract

Debugging is an essential activity in software development used to identify and fix errors in programs. Traditional debuggers such as GDB provide powerful features, but understanding the underlying operating system concepts behind debugging is equally important. This project aims to design and implement a lightweight Linux-based debugger using C programming and Linux system calls. The debugger will allow users to execute programs under control, set breakpoints, inspect registers, view memory contents, and trace system calls.

The project uses Linux APIs such as `fork()`, `exec()`, `wait()`, and `ptrace()` to monitor and control child processes. Concepts like process management, inter-process communication, signals, and system programming are applied to implement debugging functionalities. The debugger will provide a command-line interface where users can issue commands to start, pause, continue, and inspect a target program.

The expected outcome is a functional mini-debugger that demonstrates how operating system mechanisms support debugging tools. This project helps students gain practical knowledge of Linux internals, process control, and system programming concepts while developing a useful software tool.

2. Problem Statement

Software developers often face difficulties in identifying program errors, memory issues, and unexpected behavior during execution. Existing debugging tools are complex for beginners and hide the

internal operating system mechanisms involved. This project aims to develop a simple Linux-based debugger that helps users understand and analyze program execution using system calls and process control techniques. It demonstrates the relevance of operating systems concepts such as process management, signals, and system-level debugging.

3. Objectives

1. To develop a command-line debugger for Linux systems.
2. To implement breakpoints and process control features.
3. To inspect CPU registers and memory contents.
4. To understand Linux system calls and OS concepts related to debugging

4. Proposed Methodology

1. Use fork() and exec() to create and execute target processes.
2. Use ptrace() to monitor and control program execution.
3. Implement breakpoint handling and register inspection.
4. Use wait() and signal handling for process synchronization.
5. Provide user commands for debugging operations.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	Create and manage processes
fork()	Create child process
exec()	Execute target program
wait()	Synchronize parent and child
ptrace()	Debug and trace processes
Signals	Handle process events

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030418	A. Charan Teja	Process creation and execution
2520030419	S. Teja	Debugging and ptrace() implementation
2520030475	E. Ashwith Reddy	User interface, testing, documentation & GitHub

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Development platform
C / C++	Implementation
GCC	Compilation
GDB	Reference debugger
Git & GitHub	Version control

8. Expected Outcome

The project will produce a lightweight debugger capable of controlling processes, setting breakpoints, viewing registers, and tracing program execution. It will demonstrate practical applications of operating systems concepts and Linux system programming techniques.

This topic is highly suitable for an OSSP project because it directly uses **process management, system calls, signals, memory management, and Linux APIs**, which are core Operating Systems concepts

9. GitHub Repository

GitHub Repository Link: https://github.com/tejashetteni2767-svg/Team-19_OSSP

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____